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[MANDATORY NOTICE]

SYNTHETIC DEMONSTRATION DATA

This document is fictional test data created for the VYASA prototype.

It is not an actual operating procedure, engineering standard,
regulatory requirement, or safety instruction.

All numerical values are example synthetic demonstration values only.

SECTION 1: EQUIPMENT TAGGING CONVENTIONS & CODING STANDARDS

1.1 Tag Prefix Conventions (Synthetic Demonstration Reference)

- P: Pump equipment identifier (e.g., P-101, P-102 in hydrocarbon service).
- C: Compressor unit identifier (e.g., C-201, C-202 in recycle gas service).
- V: Pressure vessel / drum identifier (e.g., V-301 suction drum, V-305 discharge separator).
- FS: Flare stack identifier (e.g., FS-501 elevated emergency flare system).
- MOV: Motor-operated valve.
- PSV: Pressure safety valve.

1.2 Drawing Numbering System

Drawings in this reference follow the synthetic standard PID-MRPL-SYN-001 through PID-MRPL-SYN-004.

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SECTION 2: STANDARD EQUIPMENT & INSTRUMENT SYMBOLS

2.1 Pump & Compressor Symbols

- Centrifugal Pump Symbol: Represented in standard schematics by a circle with an inscribed tangential discharge triangle pointing towards the discharge line (or represented as a blue square in simplified schematic audits).
- Centrifugal Compressor Symbol: Represented by a trapezoid showing flow entering the larger base and exiting the narrower top, indicating gas compression.

2.2 Valve Symbols

- Standard Gate/Globe Valve Symbol: A valve is represented by two opposing triangles meeting at a common apex or central point (commonly known as a bowtie or hourglass symbol, or represented as a red circle in simplified audit diagrams).
- Check Valve: Opposing triangles with an internal directional check arrow.
- Control Valve: Opposing triangles surmounted by an actuator dome or diaphragm.

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SECTION 3: FLARE SYMBOLS & EQUIPMENT TAG RELATIONSHIPS

3.1 Flare Stack Symbols

- Flare Stack (FS-501): Represented by a vertical continuous line terminating with an open flare burner tip and flame icon.
- Knockout Drum (V-308): Vertical cylindrical vessel with inlet deflector baffle and mist eliminator pad.

3.2 Example Equipment Tag Relationships

- Pump P-102 suction line connects directly from Feed Vessel V-301.
- Compressor C-202 discharge header feeds directly into High-Pressure Separator V-305.
- Emergency relief lines from V-301, V-305, and Compressor C-202 route into the main flare header terminating at Flare Stack FS-501.